

Chapter 3

Numerical differentiation

3.1 Introduction

Everyone who possesses a car and/or a driver's licence is familiar with speeding tickets. In The Netherlands, speeding tickets are usually processed in a fully automated fashion, and the perpetrator will receive the tickets within a couple of weeks after the offence. The Dutch police optimized the procedures of speed control such that this effort has become very profitable to the Dutch government. Various strategies for speed control are carried out by police forces, which are all based on the position of the vehicle at consecutive times. The actual velocity follows from the first-order derivative of the position of the vehicle with respect to time. Since no explicit formula for this position is available, the velocity can only be estimated using an approximation of the velocity based on several discrete vehicle positions at discrete times. This motivates the use of approximate derivatives, also called *numerical derivatives*. If the police want to know whether the offender drove faster before speed detection (in other words, whether the perpetrator hit the brakes after having seen the police patrol), or whether the driver was already accelerating, then they are also interested in the acceleration of the 'bad guy'. This acceleration can be estimated using numerical approximations of the second-order derivative of the car position with respect to time.

Since the time-interval of recording is nonzero, the velocity is not determined exactly in general. In this chapter, the resulting error, referred to as the *truncation error*, is estimated using Taylor series. In most cases, the truncation error increases with an increasing size of the recording interval (Sections 3.2 and 3.4). Next to the truncation error, the measurement of the position of the vehicle is also prone to measurement errors. Issues that influence the results are, for example, parallax, the measurement equipment, and in some cases even the performance of the police officer (in car-videoing and laser control). These measurement errors provide an additional deterioration of the approximation of the speed and acceleration. The impact of measurement errors on approximations of derivatives is treated in Section 3.3.

3.2 Simple difference formulae for the first derivative

Suppose f is a continuously differentiable function. The *forward difference* is defined as

$$Q_f(h) = \frac{f(x+h) - f(x)}{h}, \quad h > 0,$$

in which h is called the *step size*. By definition,

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x),$$